

Project overview · for the team

DBARS : Demand Based Bus Allocation & Routing System · BMTC, Bengaluru

We take BMTC's published timetable and build three things from it: a journey planner for passengers, a planning tool for depots, and an open data feed anyone can use. One dataset, three audiences.

6,737 Route directions , **3,940** Bus numbers , **4,883** Stops , **44** Depots planned , **15,636** Crew duties

What it does

Find the bus : For passengers

Type where you are and where you're going. It finds direct buses, and journeys with one or two changes when there's no direct one. Buy a ticket with a QR code, see the route drawn on a map, and check which Metro station is nearby — and which bus gets you there.

Plan the fleet : For depot managers

The part no travel app does. It works out the smallest number of buses that can actually run the timetable, how many kilometres they run empty between trips, and how many crew duties are needed to staff them. Change a timetable and see what it costs in buses.

Open the data : For everyone else

The whole network is exported as GTFS — the standard format journey planners read. That means any other app could use BMTC's timetable without us building anything for them. Live service alerts go out in the same standard.

What makes it different

It answers both sides of the same question.

Google Maps and the Namma BMTC app tell *you* which bus to catch. Neither tells BMTC how many buses it needs to run that service. DBARS does both, from one dataset — that pairing is the whole idea.

Two honest ways to see a route.

Fewest changes and shortest journey usually aren't the same trip. Whitefield to Kengeri is 33 stops with one change, or fewer kilometres with two. We show both instead of picking for you and hiding the trade-off.

It keeps working when things break.

Route search reads files, not the database, so if MongoDB goes down people can still find their bus — only logins and tickets stop. Conductors can sell tickets with no signal at all and sync later without creating duplicates.

It says what is real and what is not.

Live bus positions are simulated by default, and the code refuses to publish them as real. Fares and crowd levels carry their own confidence notes. Nothing is dressed up as data we don't have.

It can explain itself.

A built-in assistant answers questions about routes, depots and the code itself, with the exact file and line it got the answer from. It can only read — it cannot change a ticket, a route or a plan.

WHO WOULD ACTUALLY USE IT

PERSON	WHAT THEY DO WITH IT
Passenger	Finds a bus, buys a ticket, checks how crowded it is, shares a live trip with family for safety.
Conductor	Signs on to a duty, issues tickets on board even with no network, remits cash at sign-off.
Depot manager	Sees the minimum buses and crew the timetable needs, and what a schedule change would cost.
BMTC office	Reconciles cash against digital sales, and totals the Shakti free rides the state reimburses.
Other developers	Download the GTFS feed and plug BMTC into their own app.

Shakti scheme. Karnataka gives women free bus travel. DBARS issues those tickets at zero fare with the same QR code as a paid one, and keeps a separate count of what the state owes BMTC — so free rides are still counted, not lost.

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